



The Floor is Lava: Temple Circuit

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Difficulty : Easy-Medium

Description

Deep inside an ancient temple, you discover a mystical chamber filled with N glowing pressure plates. Each plate is connected to others by fragile, crumbling stone bridges.

Once you cross a bridge, it collapses into a lava pool — no turning back.

Your mission is to step on these pressure plates, starting from a given plate S , and use every bridge exactly once, returning safely to the starting point before the lava consumes the entire floor.

Input

You will be given:

- Two integers N and S :
 - N — the number of pressure plates ($3 \leq N \leq 50$)
 - S — the starting pressure plate ($0 \leq S < N$)
- A $N \times N$ matrix G , where:
 - $G[i][j] = 1$ means there is a bridge between plate i and j
 - $G[i][j] = G[j][i]$
 - $G[i][i] = 0$



Output

- If escape is impossible, print: **NO**
- If a path exists, print: **YES**

followed by a space-separated sequence of plates, starting and ending at **S**.

Examples

Input	Output
4 0 0 1 1 0 1 0 1 1 1 1 0 1 0 1 1 0	YES 0 1 2 3 1 0